

* Calculate Information gain Weather

- Entropy of entire dataset

$$S \{ +9, -5 \} = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} \\ = 0.94$$

- Entropy of each attribute

- Entropy of Sunny

$$S_s \{ , \} = - \log_2 - - \log_2 - \\ = 0.97$$

$$\textcircled{2} (S)_c \{ , \} = - \log_2 - - \log_2 - \\ = 0$$

$$\textcircled{3} (S)_R \{ , \} = - \log_2 - - \log_2 - \\ = 0.97$$

$$\text{Information gain} = \text{Entropy (whole)} - \frac{5}{14} \text{Ent}(S) - \frac{4}{14} \text{Ent}(C) - \frac{5}{14} \text{Ent}(R)$$

$$(IG)_{\text{weather}} = 0.246$$

* Information gain for Temp. -

• Step 1: Entropy of whole dataset = 0.94

• Entropy of all attribute

$$(S)_H \{ , \} = - \frac{\log}{2} - \frac{\log}{2} = 1.0$$

$$(S)_M \{ , \} = - \frac{\log}{2} - \frac{\log}{2} = 0.91$$

$$(S)_C \{ , \} = - \frac{\log}{2} - \frac{\log}{2} = 0.81$$

$$\text{Information gain} = 0.029$$

* Information gain for Humidity

- Entropy (whole dataset) = 0.94
- Entropy of all attributes

$$(S)_{\text{High}} \{ , \} = - \frac{\log_2}{2} - \frac{\log_2}{2} = 0.98$$

$$(S)_{\text{Normal}} \{ , \} = - \frac{\log_2}{2} - \frac{\log_2}{2} = 0.59$$

$$\text{Information gain} = 0.15$$

* Information gain of wind

- Entropy (dataset)^{whole} = 0.94

- Entropy of attributes

$$(S)_{\text{strong}} \{ \} = - \frac{\log_2}{2} - \frac{\log_2}{2} = 1.0$$

$$(S)_{\text{Normal}} \{ \} = - \frac{\log_2}{2} - \frac{\log_2}{2} = 0.81$$

$$\text{Information gain} = 0.0478$$

$$\text{gain(weathr)} = 0.246$$

$$\text{gain(Temp)} = 0.029$$

$$\text{gain(Humidity)} = 0.15$$

$$\text{gain(Wind)} = 0.0478$$

* specially for Sunny Only

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$$\text{gain}(S_{\text{sunny}}, \text{Temp}) = 0.57$$

$$\text{gain}(S_{\text{sunny}}, \text{Humidity}) = 0.97$$

$$\text{gain}(S_{\text{sunny}}, \text{Wind}) = 0.019$$